PORTFOLIO PROJECT · SQL

Who leaves, and why?

Employee attrition analysis of IBM's HR dataset in MySQL

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1,470 employees · 35 variables · 10 queries

THE QUESTION

16.1% of employees left. Which ones, and what can HR change?

The data

IBM HR Analytics: 1,470 employees and 35 variables covering pay, role, satisfaction and whether they left.

The tool

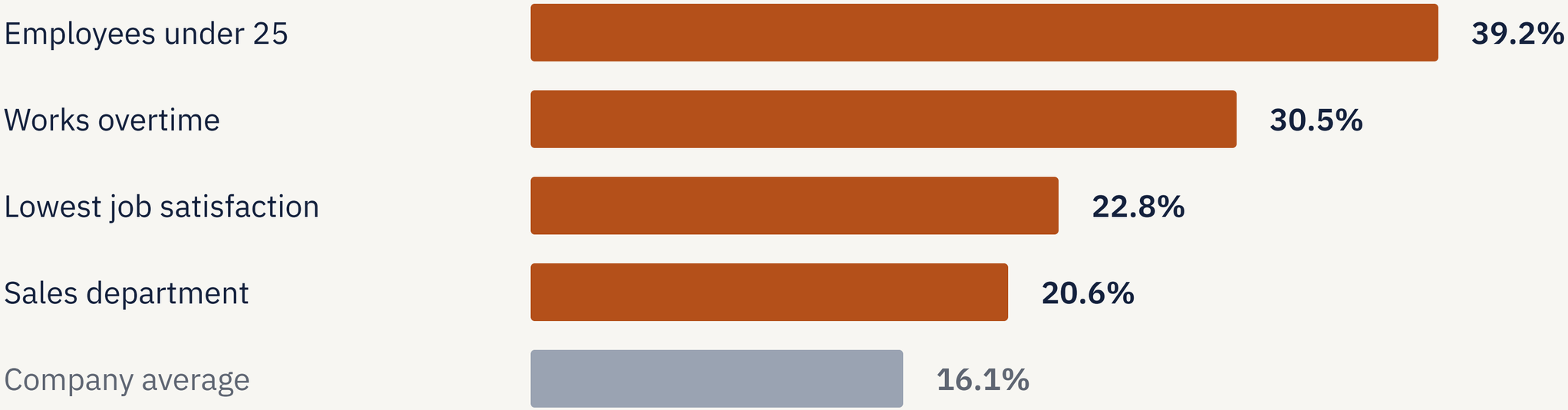
MySQL Workbench. Ten queries, each answering one business question.

The prep

No nulls or duplicates. Dropped 3 constant columns. Stripped a hidden UTF-8 marker in Python that broke the import.

ATTRITION RATE BY SEGMENT

Four groups leave far above average



Groups overlap: one employee can fall into several of them.

THE STRONGEST SINGLE DRIVER

Employees on overtime leave at nearly 3× the rate

30.5%

of overtime workers left
127 of 416

10.4%

of everyone else left
110 of 1,054

COMPENSATION

Pay is lowest where people leave

6.5×

Manager vs Sales Representative pay: $17,181 vs $2,626 a month

2.4×

Pay growth from year 0 to year 14, mostly after year 10

Job role	Avg monthly pay	Headcount
Manager	$17,181	102
Research Director	$16,033	80
Healthcare Representative	$7,528	131
Manufacturing Director	$7,295	145
Sales Executive	$6,924	326
Human Resources	$4,235	52
Research Scientist (lowest 3)	$3,239	292
Laboratory Technician (lowest 3)	$3,237	259
Sales Representative (lowest 3)	$2,626	83

WHO TO RETAIN FIRST

The flight-risk profile, in one query

Age	18 to 40
Roles	Lab Technician, Research Scientist, Sales Rep
Monthly pay	$1,600 to $2,300
Tenure	0 to 10 years
Pattern	Overtime, satisfaction of 1 or 2

Profile of employees who left while on overtime and reporting low satisfaction.

```
-- query 9 of 10
SELECT EmployeeNumber, Age, JobRole,
       MonthlyIncome, YearsAtCompany
FROM ibm_hr.HR
WHERE Attrition = 1
  AND JobSatisfaction <= 2
  AND OverTime = 1
ORDER BY MonthlyIncome ASC;
```

RECOMMENDATIONS

Six actions for HR

Now

Watch overtime in Sales

Flag sustained overtime for manager review and rebalance workloads.

Now

Raise entry-level pay

The three lowest-paid roles earn $2,600 to $3,200 a month.

Now

Onboard under-25s for 90 days

Structured check-ins for the group with 39.2% attrition.

Next quarter

Front-load raises

Move pay growth into years 1 to 5 instead of after year 10.

Next quarter

Quarterly satisfaction pulse

Score 1 leaves at 22.8%, score 4 at 11.3%. Track it by department.

Next quarter

Exit interviews in Sales

The data shows where people leave, not why.

HONEST LIMITS

What this can't say, and what I'd do next

Limits

- IBM built this dataset for teaching. It is not a real company.
- One snapshot in time: it shows association, not cause.
- Groups overlap, so single-variable rates double-count.

Next steps

- Logistic regression to rank all drivers together.
- A monthly Power BI dashboard for HR.
- Exit-interview reasons to explain the Sales numbers.